



AI-Powered Collections Strategy



Leveraging Agentic AI for Scalable, Fair, and
Effective Debt Management at Geldium



System Workflow: Inputs & Decision Logic

1 Inputs

Live feed from the Task 2 predictive model: risk probability + Low / Moderate / High tier

Underlying signals: Missed_Payments, Credit_Utilization, Credit_Score, Debt-to-Income Ratio

Refreshed continuously as new payment and account activity arrives

Account context: tenure, employment status, prior outreach / hardship history

2 Decision Logic

Combines the model's risk tier with business rules (contact-frequency caps, active hardship-plan status)

Applies the calibrated decision threshold from the Model Plan to convert probability into an action tier

Moderate/High-risk and borderline cases are routed for review rather than auto-resolved

Rules + model work together – not model output alone – to pick the next step

System Workflow: Actions & Learning



Actions

Automated SMS / email payment reminders for Low–Moderate risk accounts

Offer of a short-term hardship or debt-management plan for High-risk accounts

Escalation to a Collections agent for review and approval when required

Learning Loop

Tracks repayment rate, customer engagement, and cost of collections per action taken

Feeds outcomes back to refine model thresholds and business rules over time

Re-audited on a regular cadence for accuracy and fairness (see Guardrails)

Role of Agentic AI

Autonomous

Send automated payment reminders (SMS/email) to Low–Moderate risk accounts

Choose contact channel and timing based on past customer engagement

Adjust outreach cadence based on repayment response

Flag accounts that cross a risk threshold for the next workflow step

Human Oversight

Approve hardship or debt-management plan offers

Review High-risk and borderline / near-threshold cases

Handle customer disputes or appeals

Approve actions with regulatory impact (credit-limit or account-status changes)

Responsible AI Guardrails

1

Explainability — Every automated flag carries a SHAP-based reason code (e.g., “high utilization + 3 missed payments”) so any action can be explained to a customer or regulator in plain language.

2

Bias monitoring — The Fairness Audit from the Model Plan re-runs on a regular cadence, comparing outcomes across Location, Employment_Status, Age, and Card_Type, with alerts on disparities.

3

Human-in-the-loop — Hardship offers, credit-limit changes, and High-risk cases always route to a Collections agent for review – never fully automated.

4

Compliance & audit trail — Every decision (inputs, model score, action taken) is logged for regulatory review, and customers can appeal or request human review of any outcome.

Expected Business Impact

Business KPIs

10–15% relative reduction in 30/60/90-day delinquency (pilot target)

Lower cost-per-collection vs. manual case handling

Higher hardship-plan enrollment before accounts default

Faster time-to-outreach via real-time flagging vs. periodic manual review

Customer Outcomes

Earlier, supportive contact instead of late-stage collections pressure

Consistent, explainable treatment across customer segments

Clear channel to appeal or request a human review

Scalable support that reaches the full portfolio, not just manually reviewed accounts